

Learner worksheet — one model through the day

Keep the reference sheet at the end beside your work. Use extra paper for your answers. Predict first, then uncover supplied results. In W5 and W7 circle your route: **paper / watched demonstration / ran code**. A supplied output is a recorded author calculation, not evidence that you personally ran anything.

W0 • Draw a boundary (09:00)

In plain words. Before any formula, you decide what belongs to the sample and what belongs to its surroundings. Everything later depends on what can cross that border: heat, work or atoms. Picture: a sealed specimen inside a furnace, with heat arrows crossing its wall and no atoms crossing.

Draw a sealed specimen inside a furnace. Circle the specimen as the system and label the surroundings. Can atoms cross its boundary? Can heat? Is it closed or isolated? Write one question you hope CALPHAD might help answer. No grading.

W1 • U and H (09:15)

In plain words. Energy can enter your sample as heat or as work, and the energy stored afterwards does not remember which way it came. Enthalpy adds the pressure-volume term to that stored energy, which makes the bookkeeping easy when the pressure stays fixed. Picture: an energy account with two kinds of deposit, heat and work, and one balance.

Dive deeper (optional): the sign convention for work below. If time is short, keep the idea that U and H describe the state of the sample and heat does not.

Internal energy U is a property of the sample's state. Heat q and work are energy transfers along a process. Take work **on** the sample as positive: $\Delta U = q + w_{\text{on}}$. A sample receiving 100 J and doing 30 J of work has $\Delta U = 100 - 30 = 70$ J. Heat is not another stored state variable. [1]

Enthalpy is $H = U + pV$. Pressure times volume has energy units: $1 \text{ Pa m}^3 = 1 \text{ J}$. At constant pressure, for a closed sample with only pressure-volume work in mechanical equilibrium with that pressure, $\Delta H = q_p$. This limited relation does not define H as stored heat. [1]

Worked state: $U=999 \text{ J}$, $p=100000 \text{ Pa}$, $V=0.00001 \text{ m}^3$. Thus $pV=1 \text{ J}$ and $H=1000 \text{ J}$. This is an invented definition exercise, **not volume data for the A model**.

Pair task: (a) receive 80 J and do 10 J of work: find ΔU ; (b) for a different invented state, $U=1498 \text{ J}$, $p=100000 \text{ Pa}$, $V=0.00002 \text{ m}^3$: find pV and H. (c) Must two processes with the same ΔU have the same heat transfer? Explain.

W2 • Entropy (10:00)

In plain words. Entropy belongs to the state of your sample; it is not another name for heat. A sample may lose entropy as long as its surroundings gain more, because only the total for both can never fall. Picture: the sample and its surroundings closed together in one box, with the total never going down.

Dive deeper (optional): the reversible-heat formula. If time is short, keep the idea that S describes the state and that sample plus surroundings never lose entropy.

Entropy S has units J/K. For a reversible heat transfer at constant absolute T , $\Delta S = q_{\text{rev}}/T$. A reversible 600 J input at 300 K gives $\Delta S = 2$ J/K. Do not use actual irreversible heat divided by T as a general formula for entropy change. Entropy is a state property, not a synonym for heat. [2]

Try a reversible 900 J input at 300 K. What is ΔS ? What are the units of TS ? Discuss: when a sample loses entropy, must the second law be violated? Consider the entropy change of sample plus surroundings, with that whole isolated.

W3 • Choose the potential (10:45)

In plain words. Which energy counts as 'lowest' depends on what you hold fixed. A sample in a furnace keeps its atoms, its temperature and its pressure, and under those conditions it lowers its Gibbs energy G . Picture: the sample in a heat bath; heat may cross, atoms may not.

For simple bulk systems with component amounts fixed and no additional surface, elastic, electrical or magnetic work constraints, equilibrium minimizes F at fixed T, V and G at fixed T, p . An isolated whole at fixed U, V and component amounts maximizes S . $F = U - TS$, $G = H - TS = F + pV$. Conditions choose the comparison. [3]

Worked state: $W1$'s $U=999$ J, $H=1000$ J, now $T=300$ K, $S=10$ J/K. $TS=3000$ J; $F=-2001$ J; $G=-2000$ J. Check $G-F=pV=1$ J. Negative energy is allowed; compare consistent references and the same amounts.

Fill this map for $W1(b)$'s $U=1498$ J, $pV=2$ J at $T=300$ K, $S=4$ J/K:

Step	Your value and unit
$U + pV \rightarrow H$	_____
$T \times S$	_____
$U - TS \rightarrow F$	_____
$H - TS \rightarrow G$	_____
$G - F$, compared with pV	_____

Dive deeper (optional): the F row and choice (b) at fixed T, V . The rest of the day uses G at fixed T, p .

Choose a criterion and justify it: (a) closed bulk sample at fixed T, p ; (b) closed bulk sample at fixed T, V . Why does choosing the smallest H alone miss something? If $g=-8000$ J/mol, find total G for 2 mol; explain why comparing that directly with a one-mole total would be misleading.

W4 • Two lines, one invented component (11:30)

In plain words. Each phase of the invented element is one straight line of Gibbs energy against temperature. Where the two lines cross, solid and liquid are equally good; on either side, the lower line wins. Picture: two straight lines on one graph, the steeper one taking over above the crossing.

Our entire phase model is here. A is invented, not Ni . Closed bulk, one mole of A atoms, $p=100000$ Pa, 800–1200 K, only SOLID and LIQUID. Constant within-phase h and s imply zero heat capacity there; pressure dependence and interfaces are absent. There is no kinetics or real-material prediction. [4]

Phase	h, J/mol	s, J/(mol K)	g=h-Ts, J/mol
SOLID	1000	10	1000-10T
LIQUID	7000	16	7000-16T

Worked at 900 K: $g_S = 1000 - 9000 = -8000$ J/mol; $g_L = 7000 - 14400 = -7400$ J/mol. Solid is lower by 600 J/mol. Fill the remaining rows **before** reading answers. The saved tables in W5 and W7 repeat these values: keep those pages folded or covered until W4 is done.

T, K	gS, J/mol	gL, J/mol	Lower phase or tie
900	-8000	-7400	SOLID
1000	_____	_____	_____
1100	_____	_____	_____

Sketch both lines from this table; label axes and units. Solve $1000 - 10T = 7000 - 16T$ for their crossing. Why does the higher-entropy phase's line fall faster? Why is comparing both branches a better model check than seeing only the chosen minimum? Does equilibrium tell you a melting time?

Optional full-size source plot (the table/your sketch supplies the paper route): full labelled vector plot.

W5 • Recognize the arithmetic in code (13:15)

In plain words. The code does nothing you did not do by hand in the last section: it evaluates the same two lines at several temperatures and keeps the lower value. Reading it means matching each number in the code to a number in your table. Picture: the code and your table side by side, with arrows between matching numbers.

Dive deeper (optional): reading the code. The saved table at the end of this section repeats W4's numbers for everyone.

The following is a verbatim code cell from Lesson 1. It supplies an array of kelvin values, evaluates both expressions and prints one row per temperature, including the lower value. np names NumPy; column_stack places columns beside one another; minimum selects the lower value in each row. Read and annotate it on paper; writing a program is not an entry requirement.

```
import numpy as np

temperatures = np.array([800.0, 900.0, 1000.0, 1100.0, 1200.0])
solid = 1000.0 - 10.0 * temperatures
liquid = 7000.0 - 16.0 * temperatures
lowest = np.minimum(solid, liquid)
print(np.column_stack([temperatures, solid, liquid, lowest]))
```

Before uncovering the table, predict the 1100 K row and circle the constants that express the **model**, as distinct from the input temperatures. On paper, change the input to 950 K while keeping the model constants; predict both values. A facilitator may demonstrate the same substitution. Answers follow in the separate sheet; the complete supplied output is here for annotation:

Saved output, columns T/K, gS/(J/mol), gL/(J/mol), minimum/(J/mol); display rounded to integers:

T	gS	gL	Minimum
800	-7000	-5800	-7000
900	-8000	-7400	-8000
1000	-9000	-9000	-9000
1100	-10000	-10600	-10600
1200	-11000	-12200	-12200

Underline the row agreeing with your hand calculation. Changing h or s would change the model; changing only T asks another question of the same model.

W6 • What may an optimizer change? (14:00)

In plain words. The computer may only decide how much is solid and how much is liquid. It may not change the energies, and it may not lose atoms. The best answer sits at an end, except exactly at the crossing, where every split is equally good. Picture: a straight line from all-solid to all-liquid.

Let f_L be the liquid phase amount fraction (mol of atoms in the liquid per mol of sample) and $f_S = 1 - f_L$. Both must lie between 0 and 1. These are phase amounts divided by the total, not image-area fractions. The objective to minimize at fixed T, p is $g_{\text{mix}} = (1 - f_L)g_S + f_L g_L$ subject to $0 \leq f_L \leq 1$. This is two macroscopic phases of **one component**, not an A-B atomic solution; we neglect their interface. The model parameters stay fixed.

At 900 K, $g_{\text{mix}} = -8000 + 600f_L$ J/mol. Complete:

f_L	0	0.25	0.5	0.75	1
g_{mix} , J/mol	-8000	_____	_____	_____	-7400

Which fraction minimizes energy? At $f_L = 0.2$ the formula gives -8120 J/mol; why must an optimizer reject it even though it is lower?

Read this **argument card**, not runnable code: `c=[gS,gL]; A_eq=[[1,1]]; b_eq=[1]; bounds=[(0,1),(0,1)]`. Match each to objective, amount balance or physical bounds. The variables are $[f_S, f_L]$. SciPy's `linprog` solves this linear problem; the complete Lesson 1 implementation checks success before using a returned solution. [5]

Saved SciPy result at 900 K: fractions SOLID=1, LIQUID=0; minimum=-8000 J/mol. (Signed floating-point zero is still zero.) At 1000 K, try $f_L = 0, 0.5$ and 1 by hand. Can two solvers return different valid fractions there? State the energy and balance checks that still matter.

The advanced steps read this same problem with many candidate states instead of two.

W7 • Load, evaluate, minimize (14:45)

In plain words. A database file only stores the energy formulas; loading it computes nothing. Evaluating gives the energy of each phase, and only the minimization picks the answer, so a calculation that leaves a phase out can only end higher. Picture: three separate buttons, load, evaluate and minimize, pressed in that order.

The course's tiny TDB file stores the same expressions. These are its exact energy records, not a complete database file:

```
PARAMETER G(SOLID,A;0) 800 1000-10*T; 1200 N !
PARAMETER G(LIQUID,A;0) 800 7000-16*T; 1200 N !
```

Underline each h and coefficient of $-T$. 800 and 1200 bound this declared interval; ! ends a record. The file's other lines name the dummy component and phases. Placeholder element metadata do not provide a real material.

Follow three distinct actions with the full TDB file:

Action	API name	What it does here
Load	Database	Read the two phase models; no equilibrium yet
Evaluate	calculate	Compute a phase property, even for the higher branch
Minimize	equilibrium	Use allowed phases/components and imposed conditions

Conditions are T in K, $p=100000$ Pa, $N=1$ mol; candidates are A, SOLID and LIQUID. GM is molar Gibbs energy in J/mol, HM molar enthalpy in J/mol and SM molar entropy in J/(mol K). In this $N=1$ route phase amounts numerically equal fractions. Do not assume arbitrary total-N support or general array shapes from this example.

Saved pycalphad output from the course's own run:

T, K	SOLID GM, J/mol	LIQUID GM, J/mol	Minimum GM, J/mol	Returned used phase amount, mol
900	-8000	-7400	-8000	SOLID 1
1000	-9000	-9000	-9000	LIQUID 1
1100	-10000	-10600	-10600	LIQUID 1

For both phases at each listed T, HM is [1000,7000] J/mol and SM is [10,16] J/(mol K), in SOLID/LIQUID order. At 1000 K the returned LIQUID 1 is just one minimizer; all balanced nonnegative mixtures have the same energy.

Pair task: connect each TDB term to W4, compare both GM columns with your table, and explain why evaluating liquid at 900 K is valid. Can loading a database alone establish the equilibrium? Predict what happens if LIQUID is excluded at 1100 K, then uncover: **saved SOLID-only result = -10000 J/mol, SOLID 1**. Why is that not the unrestricted minimum? No database writing is needed today.

W8 • Consolidate and catch an error (15:30)

In plain words. A low number alone does not make an answer right. Before you trust a result, check its units, its amount basis, and that its amounts are allowed and add up. Picture: a short checklist pinned beside every answer: units, basis, amounts.

At **1050 K**, calculate both Gibbs energies, select the lower phase, write the allowed fraction range and amount balance, and predict the minimum energy.

Two fictional reports deliberately contain errors. Correct each and explain:

- A: at 900 K, " $g_S = -8$ J/mol, $g_L = -7.4$ kJ/mol, so solid wins."
- B: at 900 K, " $f_S = 0.8$, $f_L = 0.4$ satisfies the one-mole balance."

Does observing a low objective alone establish a valid equilibrium result?

W9 • A research map, not a new calculation (16:00)

In plain words. Today's invented lines answer questions about the invented element only. A real alloy and its grain boundaries need further inputs, and this section names them before anyone calculates anything. Picture: a map of three boxes, bulk composition, boundary sites and competing boundary structures, with an empty slot under each arrow.

Optional, short: Day 2 starts from this map.

A real Ni-X question adds a second element. We would need an assessed model for the chosen components, phases, temperature/composition range and conditions. A grain-boundary question also needs a boundary description, sites/area basis, exchange or closed-inventory conditions, and a model with supporting evidence. Comparing competing boundary structures needs compatible thermodynamic references and validation. Our unary bulk lines supply none of those data. [4]

Sort the cards into **known from today's model**, **model/input needed**, or **cannot conclude from today's model**, and explain your placement:

1. A's two branches cross at 1000 K.
2. A suitable Ni-X bulk Gibbs model with provenance and validity range.
3. Boundary geometry/site inventory and segregation/structural-state energies.
4. The solute concentration at an actual Ni grain boundary.
5. A real boundary changes structure at 1000 K because the A lines cross there.
6. Whether solute may exchange with a large reservoir or a finite closed sample.

Draw the question sequence: bulk composition → boundary sites/exchange → competing boundary structures. Add one missing input under each arrow. No segregation equation, binary equilibrium or real transition is calculated.

W10 • Supported exit (16:30)

In plain words. This last section shows which ideas you can already use on your own and which still need help. Use it to choose what to practise next, not to collect a score. Picture: the four exit questions as four steps you tick off, with a note beside any step where you needed a hint.

Use the reference sheet and calculator if useful, but first try without the answer sheet. Mark any hint, partner help or supplied result you use. This is a conversation about next practice, not a certification score.

1. Choose the equilibrium potential for a closed bulk sample at fixed T,p. Distinguish it from H and U; what changes if T,V are fixed instead?
2. At **975 K**, find gS and gL, identify the lower branch and explain the crossing at 1000 K. Does the crossing fix a unique phase fraction?
3. State variables, objective and amount constraint. A report labels an energy “–8 kJ” for one mole and compares it directly with “–8000 J/mol.” State the unit/basis conversion needed. Is the unit mistake evidence of a different thermodynamic model?

4. Why can these lines not predict Ni-X grain-boundary enrichment or a structural transition? Name at least one missing bulk input and one missing boundary input.

Feedback: which explanation needed help? Which condition or unit will you check first next time? Agree a continuation using the README routes.

Keep-beside-you reference sheet

Symbol / word	Meaning and units
System / closed / isolated	Chosen matter / no matter exchange / no matter or energy exchange
U	Internal energy of sample, J; $\Delta U = q + \text{work on sample}$
H	Enthalpy, $U + pV$, J; not a universal synonym for heat
S; T	Entropy, J/K; absolute temperature, K
F	Helmholtz energy, $U - TS$, J; fixed T,V comparison (some texts write A; here A is a component)
G	Gibbs energy, $H - TS$, J; fixed T,p comparison
n; $g = G/n$	Amount, mol of atoms; molar Gibbs energy, J/mol
h; s	Molar enthalpy J/mol; molar entropy J/(mol K)
fS, fL	Nonnegative phase amount fractions (not composition mole fractions), summing to one
Equilibrium	Minimum appropriate potential among allowed states under constraints; not a rate
Model / algorithm / output	Physical expressions and parameters / calculation procedure / numerical result to check
CALPHAD	CALculation of PHase Diagrams; phase models and equilibrium calculations, with model assessment requiring evidence

The potential criteria assume simple bulk conditions and fixed component amounts as in W3. Only the synthetic A model W4 is used for numerical phase comparison. Its 1000 K crossing fixes equal energies, not a unique fraction.

Sources for the condensed explanations: [1] R. Jaramillo, MIT 3.020, Lecture 3, 2021, First Law¹. [2] R. Jaramillo, MIT 3.020, Lecture 5, 2021, Entropy². [3] R. Jaramillo, MIT 3.020, Lecture 6, 2021, Thermodynamic potentials³. No DOI is listed for these lecture notes; wording/examples here are original. [4] The course's synthetic model contract defines our assumptions. [5] SciPy developers, linprog⁴. API roles are also documented by the pycalphad developers: phase evaluation⁵ and equilibrium⁶.

¹https://ocw.mit.edu/courses/3-020-thermodynamics-of-materials-spring-2021/mit3_020s21_l03.pdf

²https://ocw.mit.edu/courses/3-020-thermodynamics-of-materials-spring-2021/mit3_020s21_l05.pdf

³https://ocw.mit.edu/courses/3-020-thermodynamics-of-materials-spring-2021/mit3_020s21_l06.pdf

⁴<https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.linprog.html>

⁵<https://pycalphad.org/docs/latest/examples/LegacyEnergySurface.html>

⁶<https://pycalphad.org/docs/latest/examples/EquilibriumWithOrdering.html>